

Heap-Konstruktion

Übungsblatt mit steigender Schwierigkeit
(Automatisch generiert)

Algorithmen und Datenstrukturen I

Hinweise

Heap-Eigenschaft:

- **Max-Heap:** Jeder Knoten ist $\geq$ seinen Kindern. Wurzel = Maximum.
- **Min-Heap:** Jeder Knoten ist $\leq$ seinen Kindern. Wurzel = Minimum.

Array-Darstellung (0-indiziert):

- Wurzel bei Index 0
- Linkes Kind von Index i : Index $2i + 1$
- Rechtes Kind von Index i : Index $2i + 2$
- Eltern von Index i : Index $\lfloor (i - 1) / 2 \rfloor$

Konstruktionsmethoden:

- **Bottom-Up Heapify:** Array als Baum interpretieren, von unten nach oben heapifizieren
- **Sukzessives Einfuegen:** Elemente einzeln einfuegen und nach oben versickern (Sift-Up)

Farbmarkierungen:

- Gelb = Max-Heap Knoten
- Grün = Min-Heap Knoten
- Rot = Aktuell betrachteter Knoten
- Orange = Knoten die getauscht werden

Aufgabe 1a – Schwierigkeit: ★

Min-Heap aus aufsteigenden Zahlen

Stellen Sie die folgenden Zahlen als **Min-Heap** in Array-Darstellung dar:

1, 2, 3, 4, 5, 6

Methode: Sukzessives Einfuegen mit Sift-Up

Fragen:

1. Geben Sie den Min-Heap in Array-Darstellung an.
2. Zeichnen Sie den Min-Heap als Binaerbaum.

Loesung:

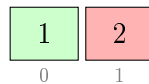
Schritt-fuer-Schritt-Konstruktion:

Schritt 0: Leerer Heap

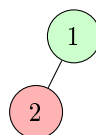
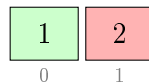
Schritt 1: Fuege 1 am Ende ein (Index 0)



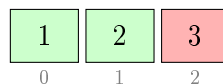
Schritt 2: Fuege 2 am Ende ein (Index 1)



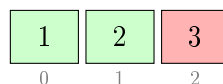
Schritt 3: $a[1]=2 \geq a[0]=1$ (Eltern): Fertig



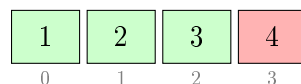
Schritt 4: Fuege 3 am Ende ein (Index 2)

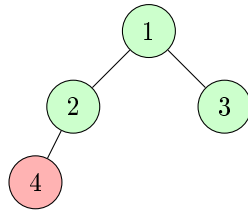


Schritt 5: $a[2]=3 \geq a[0]=1$ (Eltern): Fertig

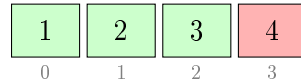


Schritt 6: Fuege 4 am Ende ein (Index 3)

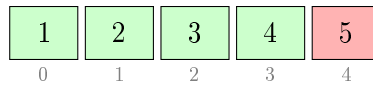




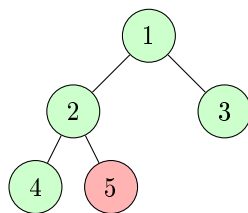
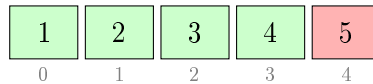
Schritt 7: $a[3]=4 \geq a[1]=2$ (Eltern): Fertig



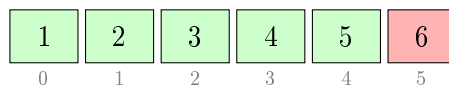
Schritt 8: Füge 5 am Ende ein (Index 4)



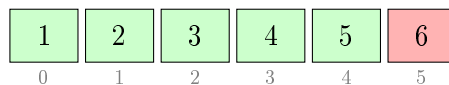
Schritt 9: $a[4]=5 \geq a[1]=2$ (Eltern): Fertig



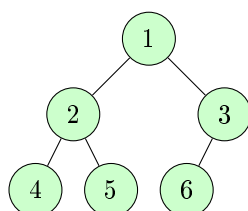
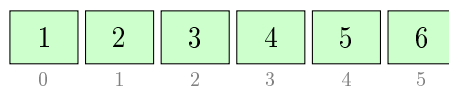
Schritt 10: Füge 6 am Ende ein (Index 5)



Schritt 11: $a[5]=6 \geq a[2]=3$ (Eltern): Fertig

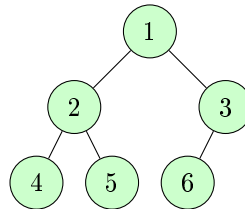


Min-Heap fertig



Antworten:

1. Min-Heap in Array-Darstellung: **[1, 2, 3, 4, 5, 6]**
2. Baumdarstellung:

**Verifikation der Heap-Eigenschaft:**

- $a[0]=1 \leq a[1]=2$ (linkes Kind) ✓
 - $a[0]=1 \leq a[2]=3$ (rechtes Kind) ✓
 - $a[1]=2 \leq a[3]=4$ (linkes Kind) ✓
 - $a[1]=2 \leq a[4]=5$ (rechtes Kind) ✓
 - $a[2]=3 \leq a[5]=6$ (linkes Kind) ✓
-

Aufgabe 1b – Schwierigkeit: ★★

Min-Heap aus absteigenden Zahlen

Stellen Sie die folgenden Zahlen als **Min-Heap** in Array-Darstellung dar:

6, 5, 4, 3, 2, 1

Methode: Bottom-Up Heapify

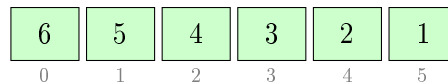
Fragen:

1. Geben Sie den Min-Heap in Array-Darstellung an.
2. Zeichnen Sie den Min-Heap als Binaerbaum.

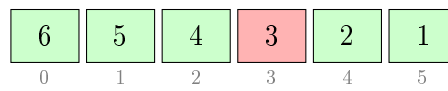
Loesung:

Schritt-fuer-Schritt-Konstruktion:

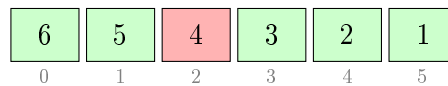
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



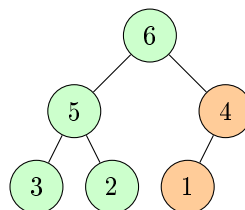
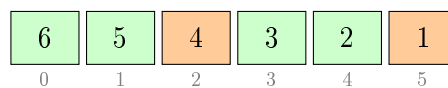
Schritt 1: Heapify ab Index 3 (Wert 3)



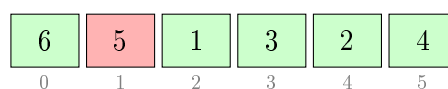
Schritt 2: Heapify ab Index 2 (Wert 4)



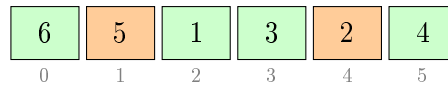
Schritt 3: $a[2]=4$ und $a[5]=1$: Tausche



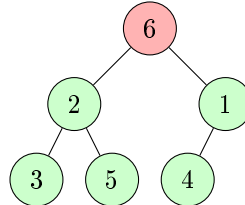
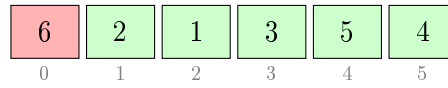
Schritt 4: Heapify ab Index 1 (Wert 5)



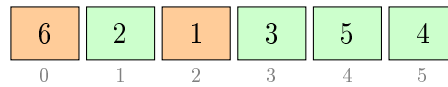
Schritt 5: $a[1]=5$ und $a[4]=2$: Tausche



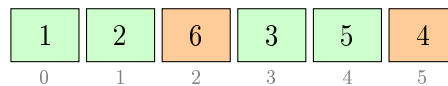
Schritt 6: Heapify ab Index 0 (Wert 6)



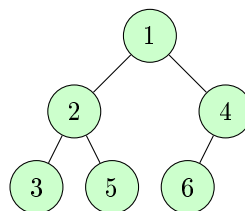
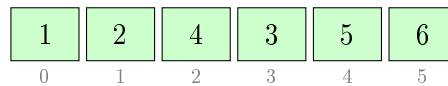
Schritt 7: $a[0]=6$ und $a[2]=1$: Tausche



Schritt 8: $a[2]=6$ und $a[5]=4$: Tausche

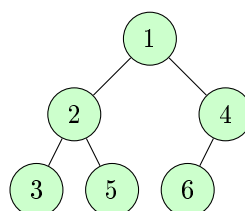


Min-Heap fertig



Antworten:

1. Min-Heap in Array-Darstellung: **[1, 2, 4, 3, 5, 6]**
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=1 \leq a[1]=2$ (linkes Kind) ✓
 - $a[0]=1 \leq a[2]=4$ (rechtes Kind) ✓
 - $a[1]=2 \leq a[3]=3$ (linkes Kind) ✓
 - $a[1]=2 \leq a[4]=5$ (rechtes Kind) ✓
 - $a[2]=4 \leq a[5]=6$ (linkes Kind) ✓
-

Aufgabe 1c – Schwierigkeit: ★★

Min-Heap aus gemischten Zahlen

Stellen Sie die folgenden Zahlen als **Min-Heap** in Array-Darstellung dar:

4, 1, 6, 2, 5, 3

Methode: Bottom-Up Heapify

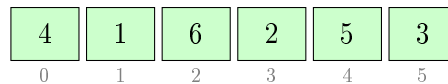
Fragen:

1. Geben Sie den Min-Heap in Array-Darstellung an.
2. Zeichnen Sie den Min-Heap als Binaerbaum.

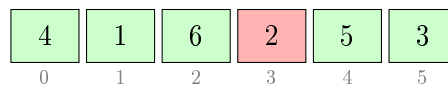
Loesung:

Schritt-fuer-Schritt-Konstruktion:

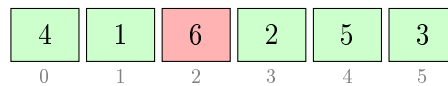
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



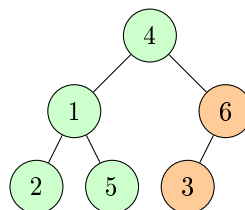
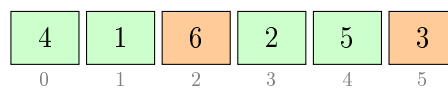
Schritt 1: Heapify ab Index 3 (Wert 2)



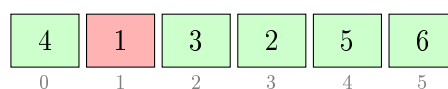
Schritt 2: Heapify ab Index 2 (Wert 6)



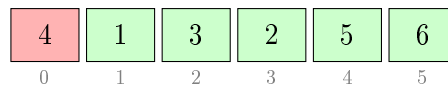
Schritt 3: $a[2]=6$ und $a[5]=3$: Tausche



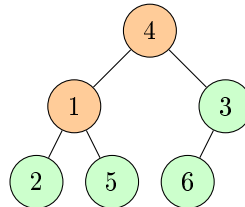
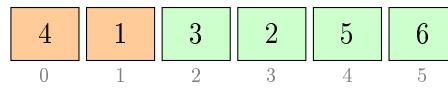
Schritt 4: Heapify ab Index 1 (Wert 1)



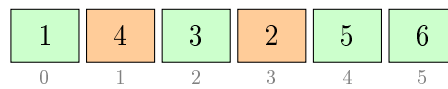
Schritt 5: Heapify ab Index 0 (Wert 4)



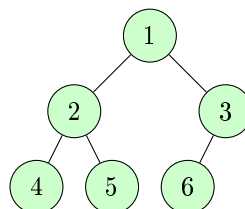
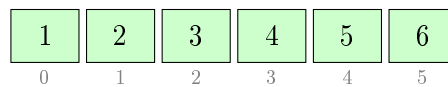
Schritt 6: $a[0]=4$ und $a[1]=1$: Tausche



Schritt 7: $a[1]=4$ und $a[3]=2$: Tausche

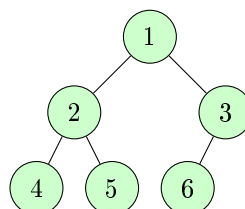


Min-Heap fertig



Antworten:

1. Min-Heap in Array-Darstellung: **[1, 2, 3, 4, 5, 6]**
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=1 \leq a[1]=2$ (linkes Kind) ✓
- $a[0]=1 \leq a[2]=3$ (rechtes Kind) ✓

- $a[1]=2 \leq a[3]=4$ (linkes Kind) ✓
 - $a[1]=2 \leq a[4]=5$ (rechtes Kind) ✓
 - $a[2]=3 \leq a[5]=6$ (linkes Kind) ✓
-

Aufgabe 1d – Schwierigkeit: ★★★

Min-Heap mit 8 Elementen

Stellen Sie die folgenden Zahlen als **Min-Heap** in Array-Darstellung dar:

7, 2, 8, 1, 5, 3, 6, 4

Methode: Bottom-Up Heapify

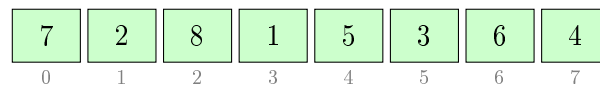
Fragen:

1. Geben Sie den Min-Heap in Array-Darstellung an.
2. Zeichnen Sie den Min-Heap als Binaerbaum.

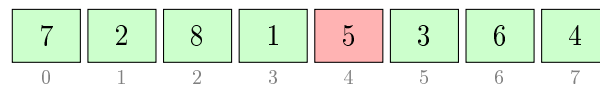
Loesung:

Schritt-fuer-Schritt-Konstruktion:

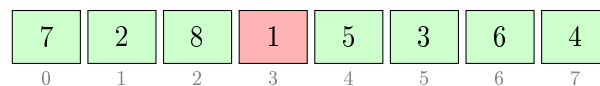
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



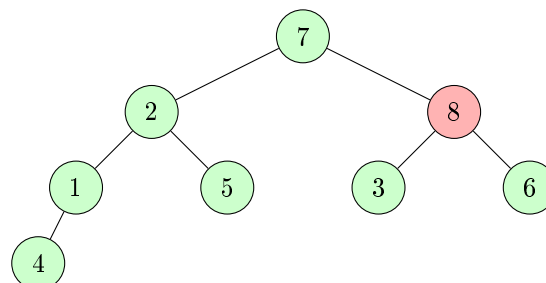
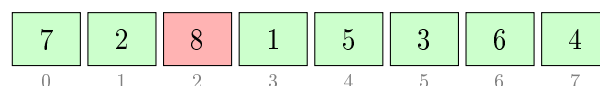
Schritt 1: Heapify ab Index 4 (Wert 5)



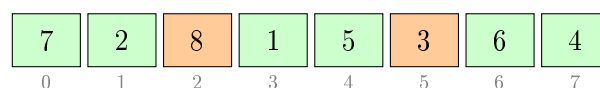
Schritt 2: Heapify ab Index 3 (Wert 1)



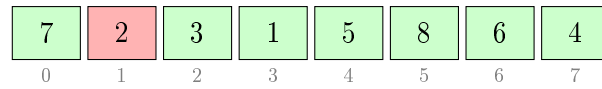
Schritt 3: Heapify ab Index 2 (Wert 8)



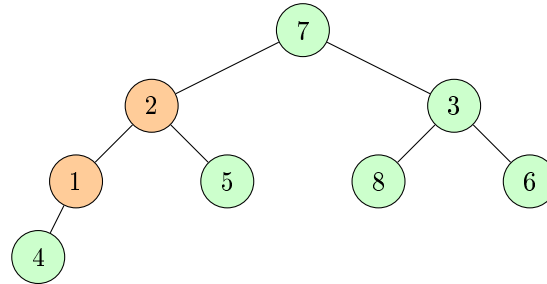
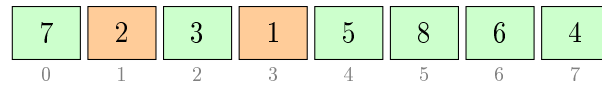
Schritt 4: $a[2]=8$ und $a[5]=3$: Tausche



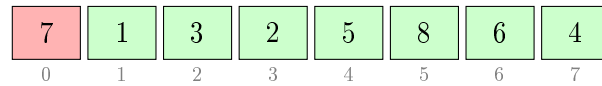
Schritt 5: Heapify ab Index 1 (Wert 2)



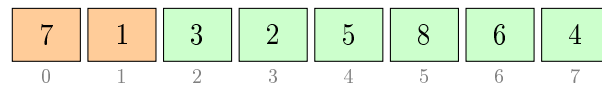
Schritt 6: $a[1]=2$ und $a[3]=1$: Tausche



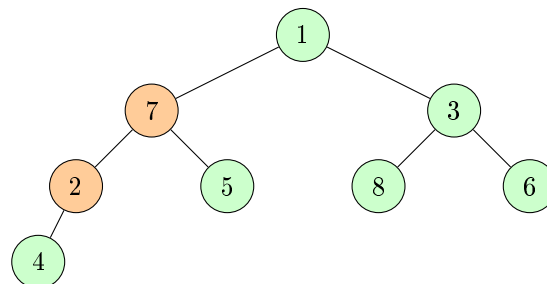
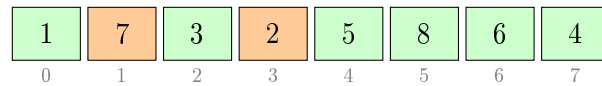
Schritt 7: Heapify ab Index 0 (Wert 7)



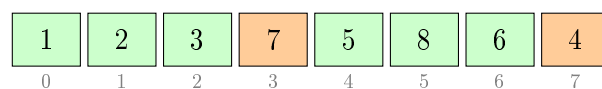
Schritt 8: $a[0]=7$ und $a[1]=1$: Tausche



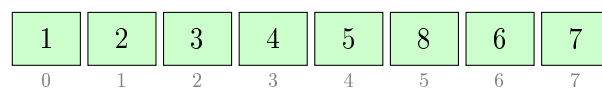
Schritt 9: $a[1]=7$ und $a[3]=2$: Tausche

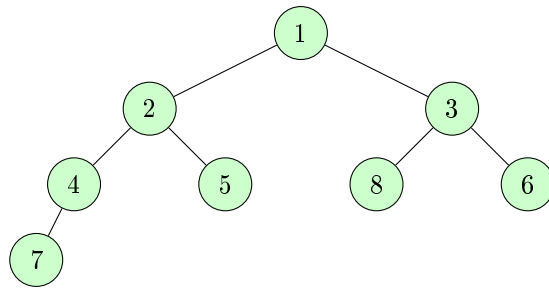


Schritt 10: $a[3]=7$ und $a[7]=4$: Tausche



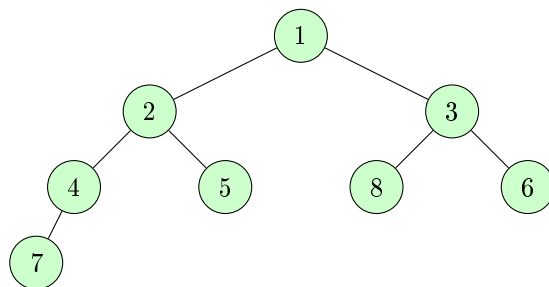
Min-Heap fertig





Antworten:

1. Min-Heap in Array-Darstellung: [1, 2, 3, 4, 5, 8, 6, 7]
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=1 \leq a[1]=2$ (linkes Kind) ✓
 - $a[0]=1 \leq a[2]=3$ (rechtes Kind) ✓
 - $a[1]=2 \leq a[3]=4$ (linkes Kind) ✓
 - $a[1]=2 \leq a[4]=5$ (rechtes Kind) ✓
 - $a[2]=3 \leq a[5]=8$ (linkes Kind) ✓
 - $a[2]=3 \leq a[6]=6$ (rechtes Kind) ✓
 - $a[3]=4 \leq a[7]=7$ (linkes Kind) ✓
-

Aufgabe 1e – Schwierigkeit: ★★★

Min-Heap durch Einfuegen

Stellen Sie die folgenden Zahlen als **Min-Heap** in Array-Darstellung dar:

5, 3, 8, 1, 4, 7, 2, 6

Methode: Sukzessives Einfuegen mit Sift-Up

Fragen:

1. Geben Sie den Min-Heap in Array-Darstellung an.
2. Zeichnen Sie den Min-Heap als Binaerbaum.

Loesung:

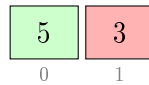
Schritt-fuer-Schritt-Konstruktion:

Schritt 0: Leerer Heap

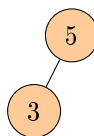
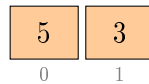
Schritt 1: Fuege 5 am Ende ein (Index 0)



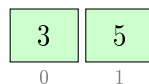
Schritt 2: Fuege 3 am Ende ein (Index 1)



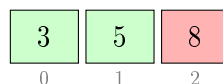
Schritt 3: $a[1]=3 < a[0]=5$ (Eltern): Tausche



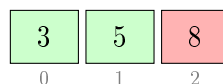
Schritt 4: 3 ist an Wurzel angekommen

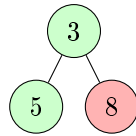


Schritt 5: Fuege 8 am Ende ein (Index 2)

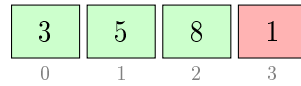


Schritt 6: $a[2]=8 \geq a[0]=3$ (Eltern): Fertig

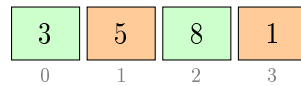




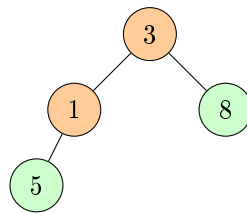
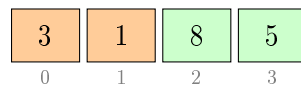
Schritt 7: Füge 1 am Ende ein (Index 3)



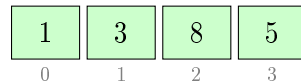
Schritt 8: $a[3]=1 < a[1]=5$ (Eltern): Tausche



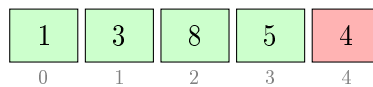
Schritt 9: $a[1]=1 < a[0]=3$ (Eltern): Tausche



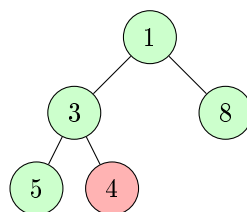
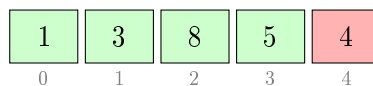
Schritt 10: 1 ist an Wurzel angekommen



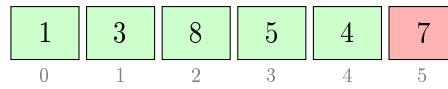
Schritt 11: Füge 4 am Ende ein (Index 4)



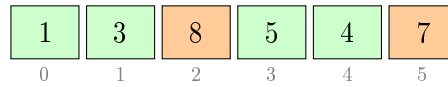
Schritt 12: $a[4]=4 \geq a[1]=3$ (Eltern): Fertig



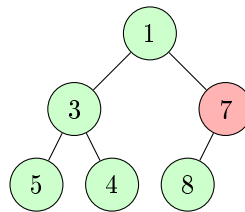
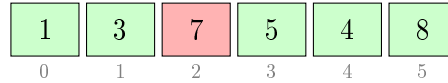
Schritt 13: Füge 7 am Ende ein (Index 5)



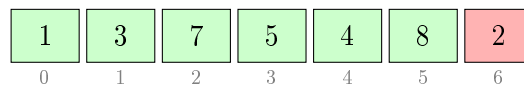
Schritt 14: $a[5]=7 < a[2]=8$ (Eltern): Tausche



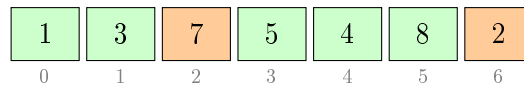
Schritt 15: $a[2]=7 \geq a[0]=1$ (Eltern): Fertig



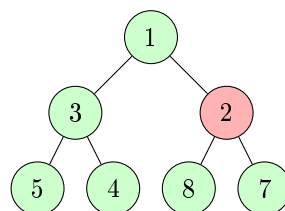
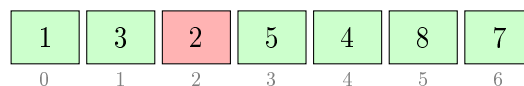
Schritt 16: Füge 2 am Ende ein (Index 6)



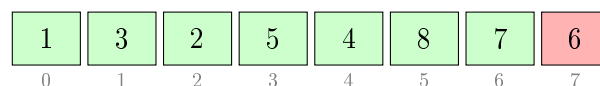
Schritt 17: $a[6]=2 < a[2]=7$ (Eltern): Tausche



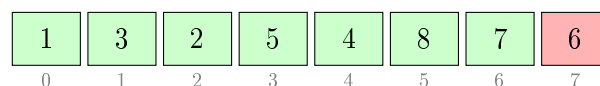
Schritt 18: $a[2]=2 \geq a[0]=1$ (Eltern): Fertig



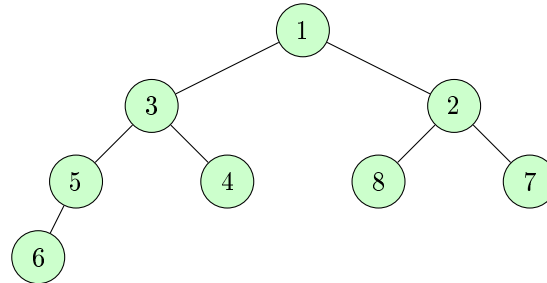
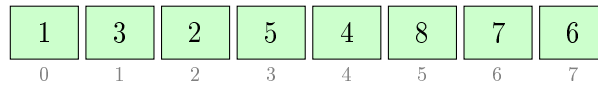
Schritt 19: Füge 6 am Ende ein (Index 7)



Schritt 20: $a[7]=6 \geq a[3]=5$ (Eltern): Fertig

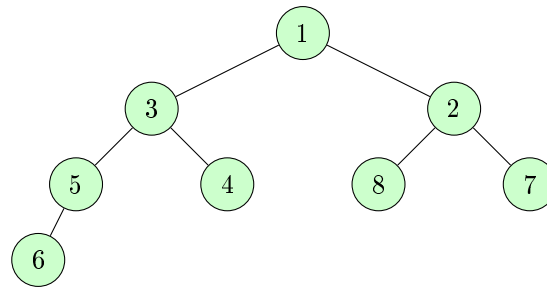


Min-Heap fertig



Antworten:

1. Min-Heap in Array-Darstellung: **[1, 3, 2, 5, 4, 8, 7, 6]**
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=1 \leq a[1]=3$ (linkes Kind) ✓
- $a[0]=1 \leq a[2]=2$ (rechtes Kind) ✓
- $a[1]=3 \leq a[3]=5$ (linkes Kind) ✓
- $a[1]=3 \leq a[4]=4$ (rechtes Kind) ✓
- $a[2]=2 \leq a[5]=8$ (linkes Kind) ✓
- $a[2]=2 \leq a[6]=7$ (rechtes Kind) ✓
- $a[3]=5 \leq a[7]=6$ (linkes Kind) ✓

Aufgabe 2a – Schwierigkeit: ★

Max-Heap aus aufsteigenden Zahlen

Stellen Sie die folgenden Zahlen als **Max-Heap** in Array-Darstellung dar:

1, 2, 3, 4, 5, 6

Methode: Bottom-Up Heapify

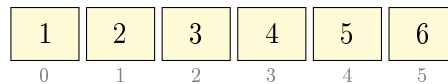
Fragen:

1. Geben Sie den Max-Heap in Array-Darstellung an.
2. Zeichnen Sie den Max-Heap als Binaerbaum.

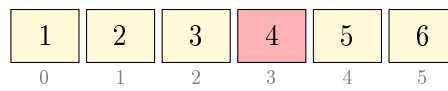
Loesung:

Schritt-fuer-Schritt-Konstruktion:

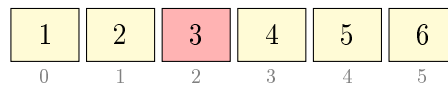
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



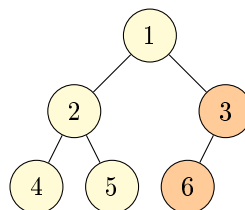
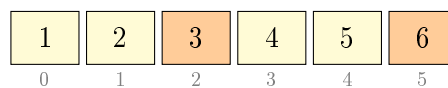
Schritt 1: Heapify ab Index 3 (Wert 4)



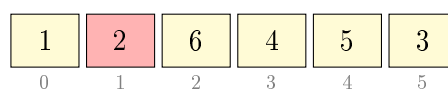
Schritt 2: Heapify ab Index 2 (Wert 3)



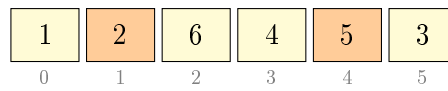
Schritt 3: $a[2]=3$ und $a[5]=6$: Tausche



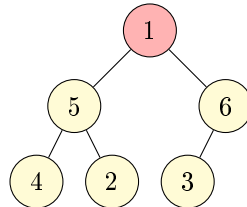
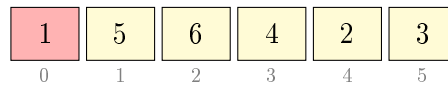
Schritt 4: Heapify ab Index 1 (Wert 2)



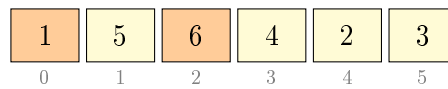
Schritt 5: $a[1]=2$ und $a[4]=5$: Tausche



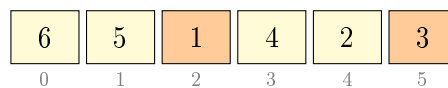
Schritt 6: Heapify ab Index 0 (Wert 1)



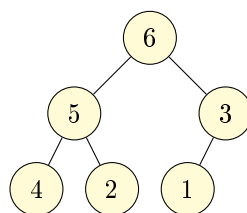
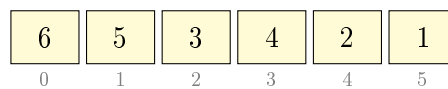
Schritt 7: $a[0]=1$ und $a[2]=6$: Tausche



Schritt 8: $a[2]=1$ und $a[5]=3$: Tausche

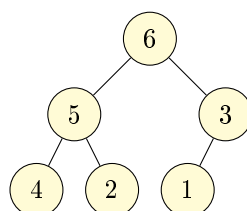


Max-Heap fertig



Antworten:

1. Max-Heap in Array-Darstellung: **[6, 5, 3, 4, 2, 1]**
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=6 \geq a[1]=5$ (linkes Kind) ✓
 - $a[0]=6 \geq a[2]=3$ (rechtes Kind) ✓
 - $a[1]=5 \geq a[3]=4$ (linkes Kind) ✓
 - $a[1]=5 \geq a[4]=2$ (rechtes Kind) ✓
 - $a[2]=3 \geq a[5]=1$ (linkes Kind) ✓
-

Aufgabe 2b – Schwierigkeit: ★

Max-Heap prüfen

Stellen Sie die folgenden Zahlen als **Max-Heap** in Array-Darstellung dar:

6, 5, 4, 3, 2, 1

Methode: Bottom-Up Heapify

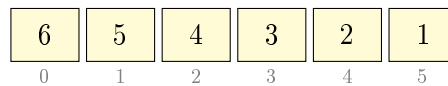
Fragen:

1. Geben Sie den Max-Heap in Array-Darstellung an.
2. Zeichnen Sie den Max-Heap als Binaerbaum.

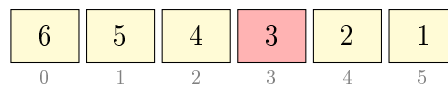
Loesung:

Schritt-fuer-Schritt-Konstruktion:

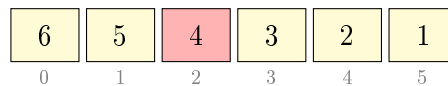
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



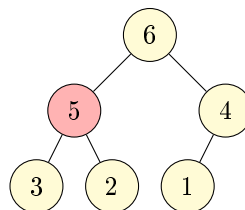
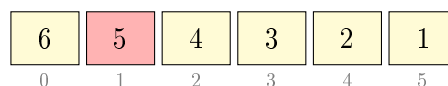
Schritt 1: Heapify ab Index 3 (Wert 3)



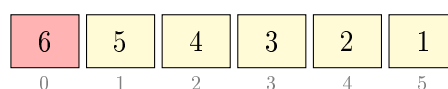
Schritt 2: Heapify ab Index 2 (Wert 4)



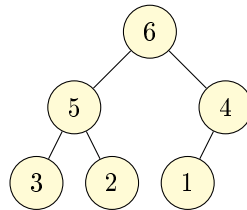
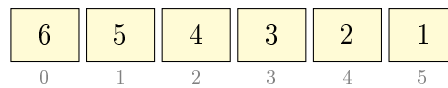
Schritt 3: Heapify ab Index 1 (Wert 5)



Schritt 4: Heapify ab Index 0 (Wert 6)

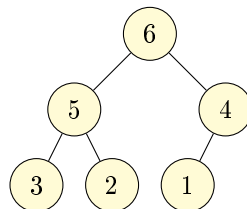


Max-Heap fertig



Antworten:

1. Max-Heap in Array-Darstellung: **[6, 5, 4, 3, 2, 1]**
2. Baumdarstellung:

**Verifikation der Heap-Eigenschaft:**

- $a[0]=6 \geq a[1]=5$ (linkes Kind) ✓
 - $a[0]=6 \geq a[2]=4$ (rechtes Kind) ✓
 - $a[1]=5 \geq a[3]=3$ (linkes Kind) ✓
 - $a[1]=5 \geq a[4]=2$ (rechtes Kind) ✓
 - $a[2]=4 \geq a[5]=1$ (linkes Kind) ✓
-

Aufgabe 2c – Schwierigkeit: ★★

Max-Heap aus gemischten Zahlen

Stellen Sie die folgenden Zahlen als **Max-Heap** in Array-Darstellung dar:

3, 6, 1, 4, 2, 5

Methode: Bottom-Up Heapify

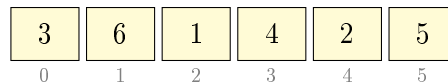
Fragen:

1. Geben Sie den Max-Heap in Array-Darstellung an.
2. Zeichnen Sie den Max-Heap als Binaerbaum.

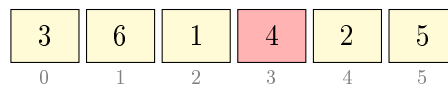
Loesung:

Schritt-fuer-Schritt-Konstruktion:

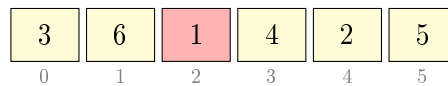
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



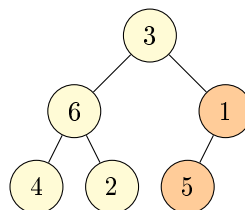
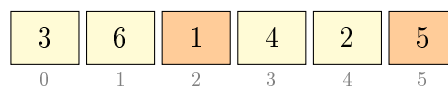
Schritt 1: Heapify ab Index 3 (Wert 4)



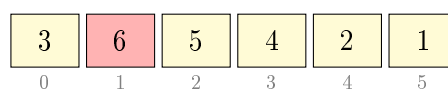
Schritt 2: Heapify ab Index 2 (Wert 1)



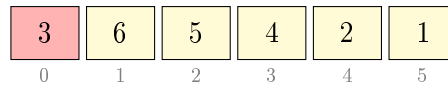
Schritt 3: $a[2]=1$ und $a[5]=5$: Tausche



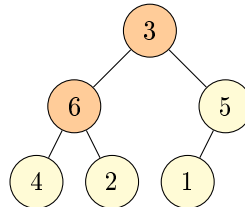
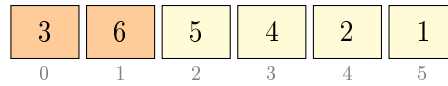
Schritt 4: Heapify ab Index 1 (Wert 6)



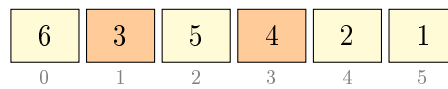
Schritt 5: Heapify ab Index 0 (Wert 3)



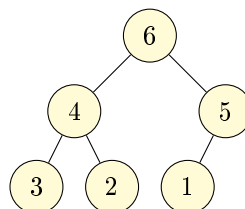
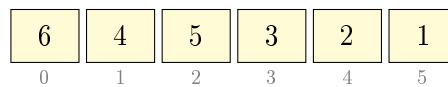
Schritt 6: $a[0]=3$ und $a[1]=6$: Tausche



Schritt 7: $a[1]=3$ und $a[3]=4$: Tausche

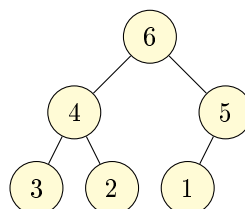


Max-Heap fertig



Antworten:

1. Max-Heap in Array-Darstellung: **[6, 4, 5, 3, 2, 1]**
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=6 \geq a[1]=4$ (linkes Kind) ✓
- $a[0]=6 \geq a[2]=5$ (rechtes Kind) ✓

- $a[1]=4 \geq a[3]=3$ (linkes Kind) ✓
 - $a[1]=4 \geq a[4]=2$ (rechtes Kind) ✓
 - $a[2]=5 \geq a[5]=1$ (linkes Kind) ✓
-

Aufgabe 2d – Schwierigkeit: ★★★

Max-Heap mit 8 Elementen

Stellen Sie die folgenden Zahlen als **Max-Heap** in Array-Darstellung dar:

4, 8, 2, 6, 1, 7, 3, 5

Methode: Bottom-Up Heapify

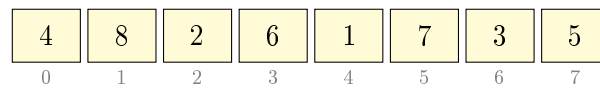
Fragen:

1. Geben Sie den Max-Heap in Array-Darstellung an.
2. Zeichnen Sie den Max-Heap als Binaerbaum.

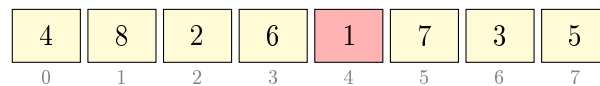
Loesung:

Schritt-fuer-Schritt-Konstruktion:

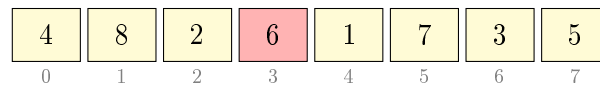
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



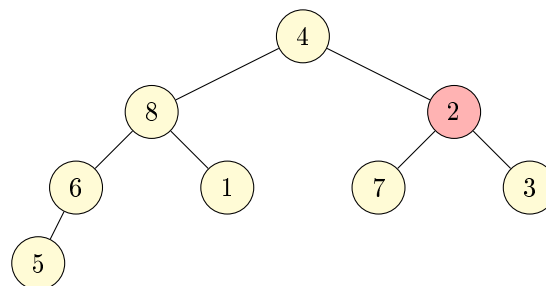
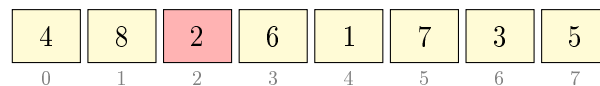
Schritt 1: Heapify ab Index 4 (Wert 1)



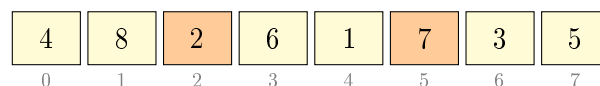
Schritt 2: Heapify ab Index 3 (Wert 6)



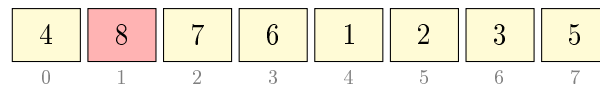
Schritt 3: Heapify ab Index 2 (Wert 2)



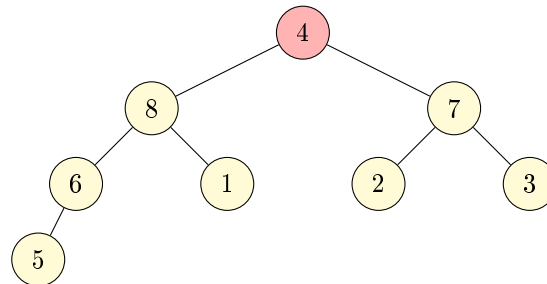
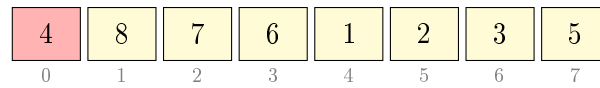
Schritt 4: $a[2]=2$ und $a[5]=7$: Tausche



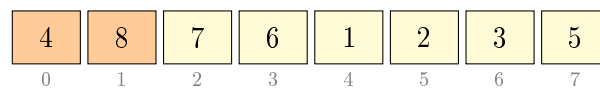
Schritt 5: Heapify ab Index 1 (Wert 8)



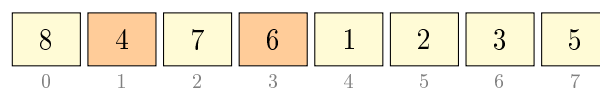
Schritt 6: Heapify ab Index 0 (Wert 4)



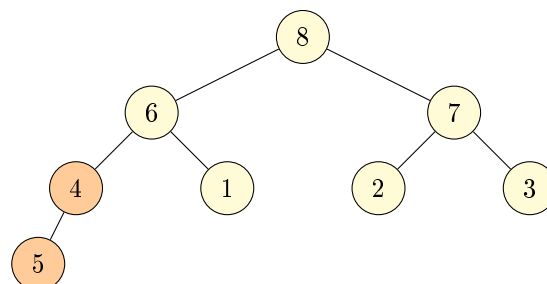
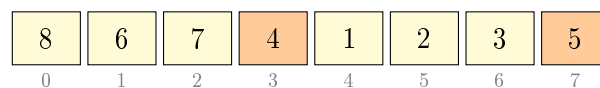
Schritt 7: $a[0]=4$ und $a[1]=8$: Tausche



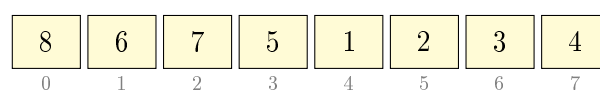
Schritt 8: $a[1]=4$ und $a[3]=6$: Tausche

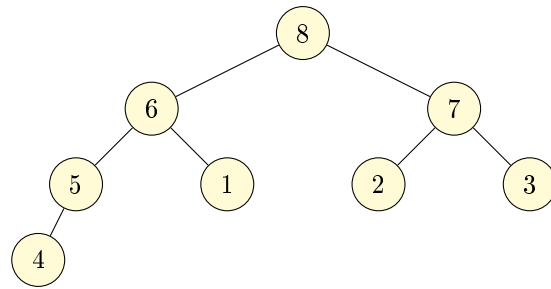


Schritt 9: $a[3]=4$ und $a[7]=5$: Tausche



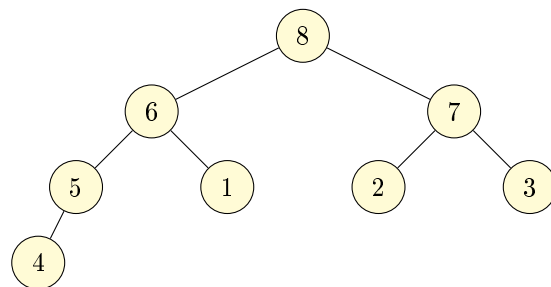
Max-Heap fertig





Antworten:

1. Max-Heap in Array-Darstellung: [8, 6, 7, 5, 1, 2, 3, 4]
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=8 \geq a[1]=6$ (linkes Kind) ✓
 - $a[0]=8 \geq a[2]=7$ (rechtes Kind) ✓
 - $a[1]=6 \geq a[3]=5$ (linkes Kind) ✓
 - $a[1]=6 \geq a[4]=1$ (rechtes Kind) ✓
 - $a[2]=7 \geq a[5]=2$ (linkes Kind) ✓
 - $a[2]=7 \geq a[6]=3$ (rechtes Kind) ✓
 - $a[3]=5 \geq a[7]=4$ (linkes Kind) ✓
-

Aufgabe 2e – Schwierigkeit: ★★★

Max-Heap durch Einfuegen

Stellen Sie die folgenden Zahlen als **Max-Heap** in Array-Darstellung dar:

3, 7, 1, 5, 2, 8, 4, 6

Methode: Sukzessives Einfuegen mit Sift-Up

Fragen:

1. Geben Sie den Max-Heap in Array-Darstellung an.
2. Zeichnen Sie den Max-Heap als Binaerbaum.

Loesung:

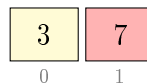
Schritt-fuer-Schritt-Konstruktion:

Schritt 0: Leerer Heap

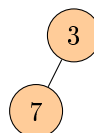
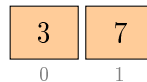
Schritt 1: Fuege 3 am Ende ein (Index 0)



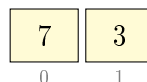
Schritt 2: Fuege 7 am Ende ein (Index 1)



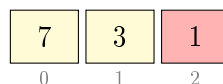
Schritt 3: $a[1]=7 > a[0]=3$ (Eltern): Tausche



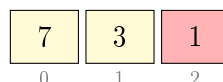
Schritt 4: 7 ist an Wurzel angekommen

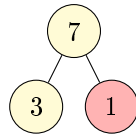


Schritt 5: Fuege 1 am Ende ein (Index 2)

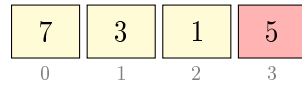


Schritt 6: $a[2]=1 \leq a[0]=7$ (Eltern): Fertig

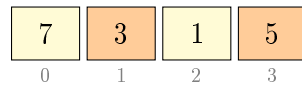




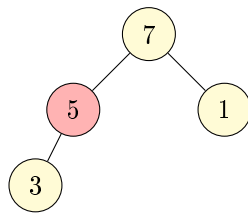
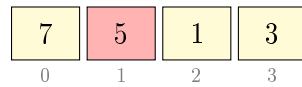
Schritt 7: Füge 5 am Ende ein (Index 3)



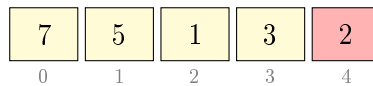
Schritt 8: $a[3]=5 > a[1]=3$ (Eltern): Tausche



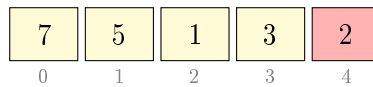
Schritt 9: $a[1]=5 \leq a[0]=7$ (Eltern): Fertig



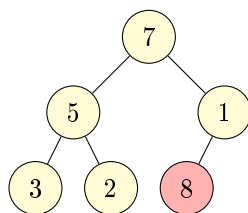
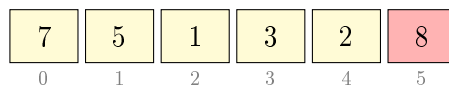
Schritt 10: Füge 2 am Ende ein (Index 4)



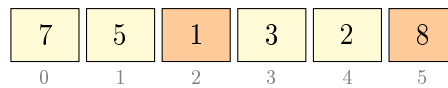
Schritt 11: $a[4]=2 \leq a[1]=5$ (Eltern): Fertig



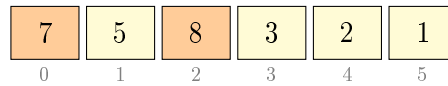
Schritt 12: Füge 8 am Ende ein (Index 5)



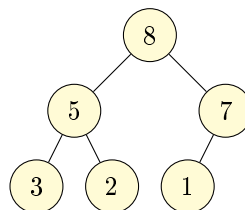
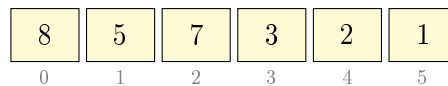
Schritt 13: $a[5]=8 > a[2]=1$ (Eltern): Tausche



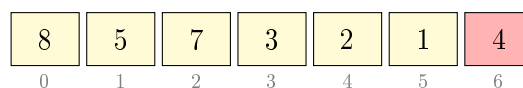
Schritt 14: $a[2]=8 > a[0]=7$ (Eltern): Tausche



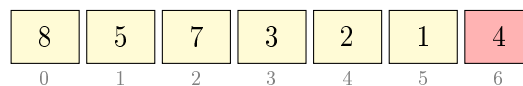
Schritt 15: 8 ist an Wurzel angekommen



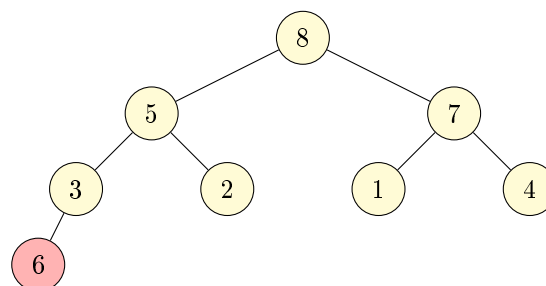
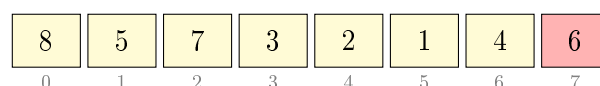
Schritt 16: Füge 4 am Ende ein (Index 6)



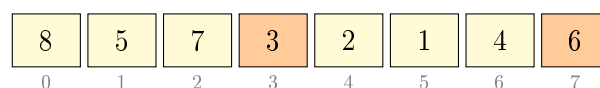
Schritt 17: $a[6]=4 \leq a[2]=7$ (Eltern): Fertig



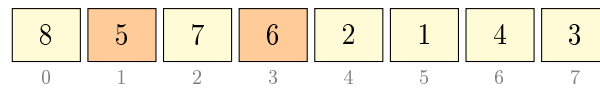
Schritt 18: Füge 6 am Ende ein (Index 7)



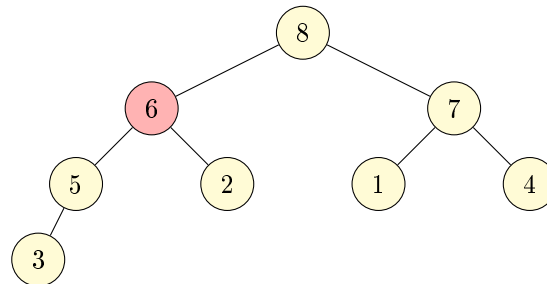
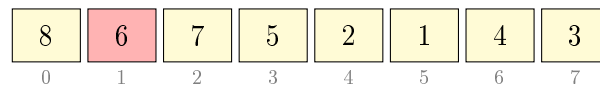
Schritt 19: $a[7]=6 > a[3]=3$ (Eltern): Tausche



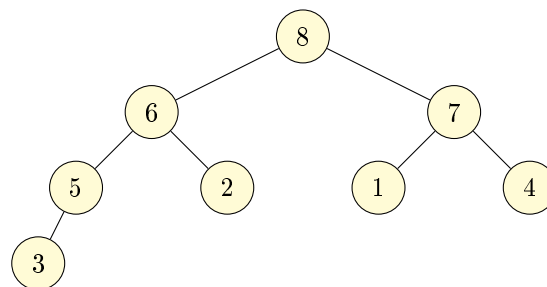
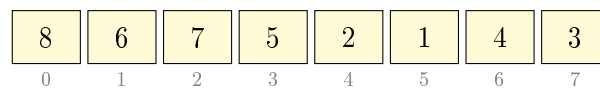
Schritt 20: $a[3]=6 > a[1]=5$ (Eltern): Tausche



Schritt 21: $a[1]=6 \leq a[0]=8$ (Eltern): Fertig

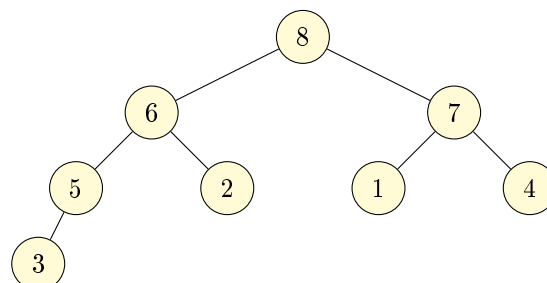


Max-Heap fertig



Antworten:

1. Max-Heap in Array-Darstellung: **[8, 6, 7, 5, 2, 1, 4, 3]**
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=8 \geq a[1]=6$ (linkes Kind) ✓
- $a[0]=8 \geq a[2]=7$ (rechtes Kind) ✓

- $a[1]=6 \geq a[3]=5$ (linkes Kind) ✓
 - $a[1]=6 \geq a[4]=2$ (rechtes Kind) ✓
 - $a[2]=7 \geq a[5]=1$ (linkes Kind) ✓
 - $a[2]=7 \geq a[6]=4$ (rechtes Kind) ✓
 - $a[3]=5 \geq a[7]=3$ (linkes Kind) ✓
-

Aufgabe 3a – Schwierigkeit: ★★★★★

Min-Heap mit 10 Elementen

Stellen Sie die folgenden Zahlen als **Min-Heap** in Array-Darstellung dar:

8, 3, 10, 1, 6, 9, 2, 7, 4, 5

Methode: Bottom-Up Heapify

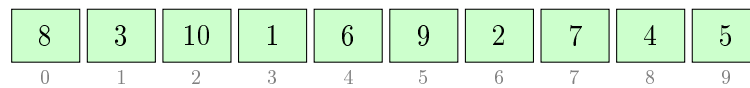
Fragen:

1. Geben Sie den Min-Heap in Array-Darstellung an.
2. Zeichnen Sie den Min-Heap als Binaerbaum.

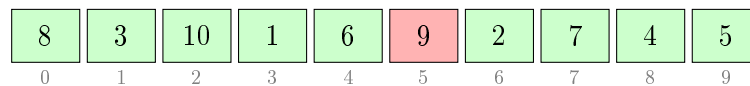
Loesung:

Schritt-fuer-Schritt-Konstruktion:

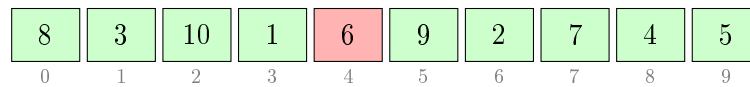
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



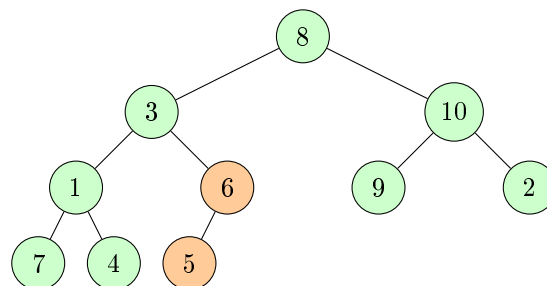
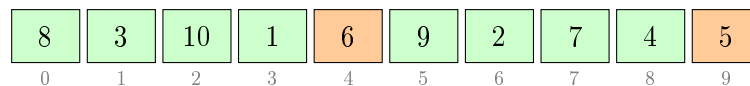
Schritt 1: Heapify ab Index 5 (Wert 9)



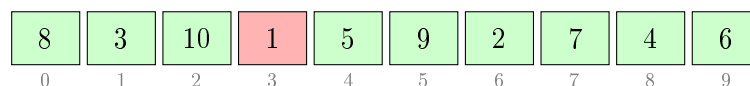
Schritt 2: Heapify ab Index 4 (Wert 6)



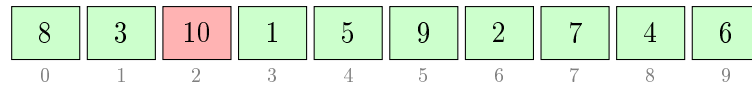
Schritt 3: $a[4]=6$ und $a[9]=5$: Tausche



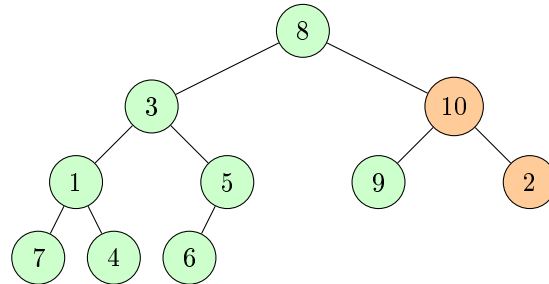
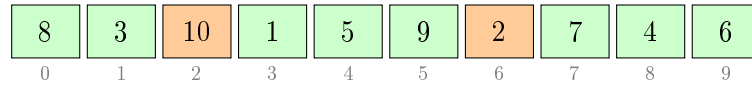
Schritt 4: Heapify ab Index 3 (Wert 1)



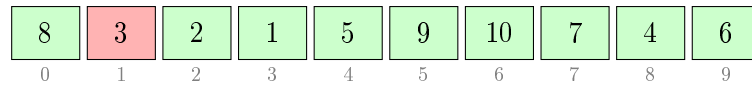
Schritt 5: Heapify ab Index 2 (Wert 10)



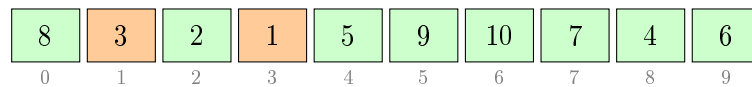
Schritt 6: $a[2]=10$ und $a[6]=2$: Tausche



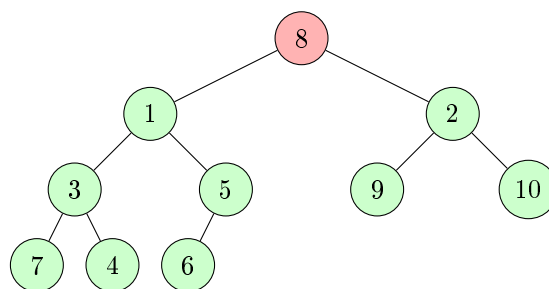
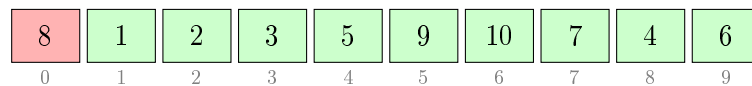
Schritt 7: Heapify ab Index 1 (Wert 3)



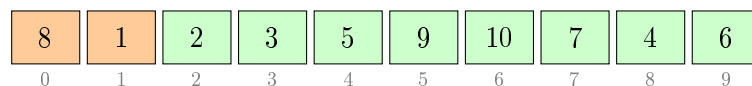
Schritt 8: $a[1]=3$ und $a[3]=1$: Tausche



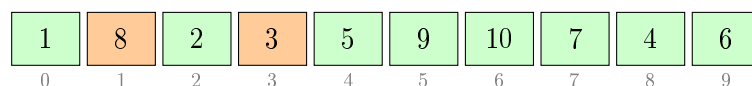
Schritt 9: Heapify ab Index 0 (Wert 8)



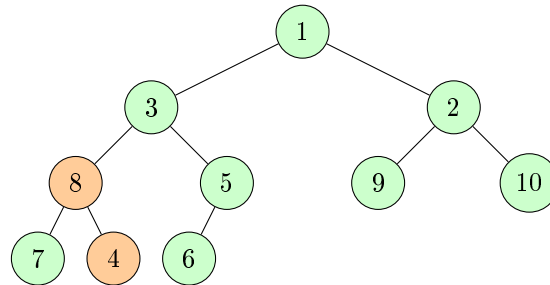
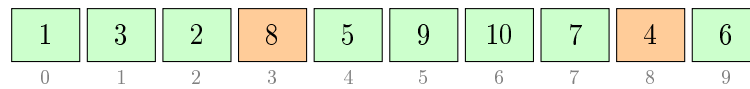
Schritt 10: $a[0]=8$ und $a[1]=1$: Tausche



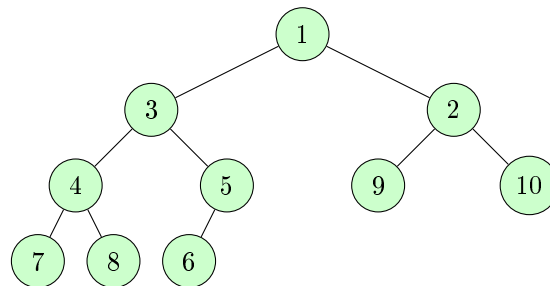
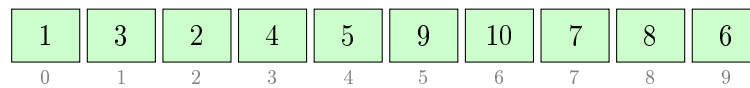
Schritt 11: $a[1]=8$ und $a[3]=3$: Tausche



Schritt 12: $a[3]=8$ und $a[8]=4$: Tausche

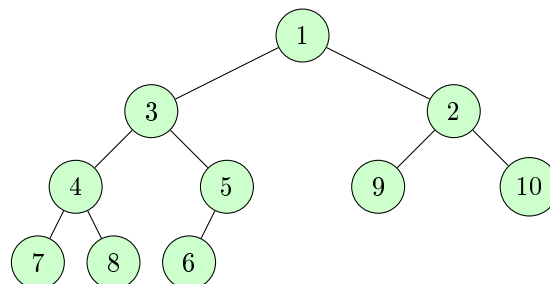


Min-Heap fertig



Antworten:

1. Min-Heap in Array-Darstellung: $[1, 3, 2, 4, 5, 9, 10, 7, 8, 6]$
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=1 \leq a[1]=3$ (linkes Kind) ✓
- $a[0]=1 \leq a[2]=2$ (rechtes Kind) ✓
- $a[1]=3 \leq a[3]=4$ (linkes Kind) ✓
- $a[1]=3 \leq a[4]=5$ (rechtes Kind) ✓

- $a[2]=2 \leq a[5]=9$ (linkes Kind) ✓
 - $a[2]=2 \leq a[6]=10$ (rechtes Kind) ✓
 - $a[3]=4 \leq a[7]=7$ (linkes Kind) ✓
 - $a[3]=4 \leq a[8]=8$ (rechtes Kind) ✓
 - $a[4]=5 \leq a[9]=6$ (linkes Kind) ✓
-

Aufgabe 3b – Schwierigkeit: ★★★★★

Max-Heap mit 10 Elementen

Stellen Sie die folgenden Zahlen als **Max-Heap** in Array-Darstellung dar:

5, 9, 2, 7, 1, 10, 4, 8, 3, 6

Methode: Bottom-Up Heapify

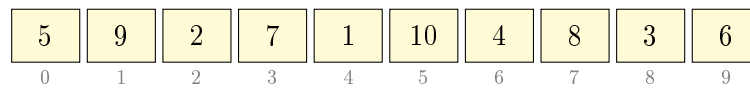
Fragen:

1. Geben Sie den Max-Heap in Array-Darstellung an.
2. Zeichnen Sie den Max-Heap als Binaerbaum.

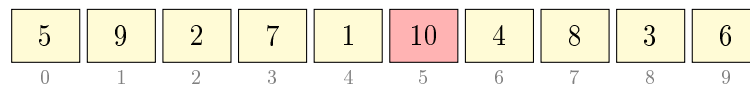
Loesung:

Schritt-fuer-Schritt-Konstruktion:

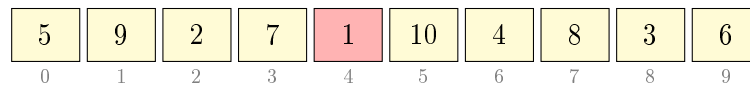
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



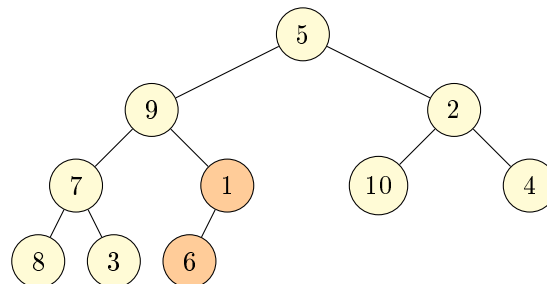
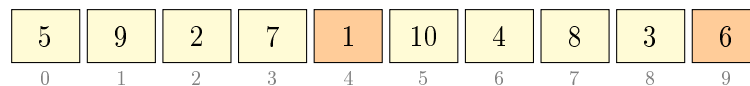
Schritt 1: Heapify ab Index 5 (Wert 10)



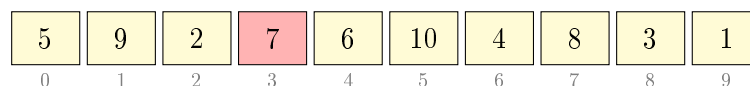
Schritt 2: Heapify ab Index 4 (Wert 1)



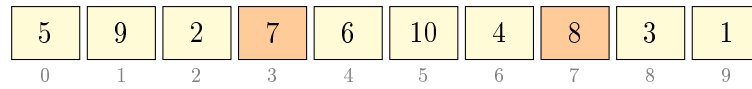
Schritt 3: $a[4]=1$ und $a[9]=6$: Tausche



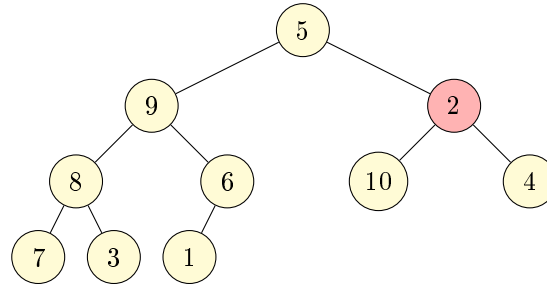
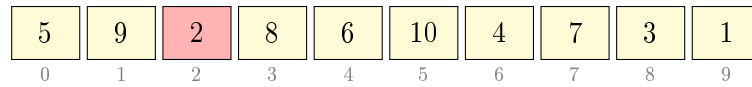
Schritt 4: Heapify ab Index 3 (Wert 7)



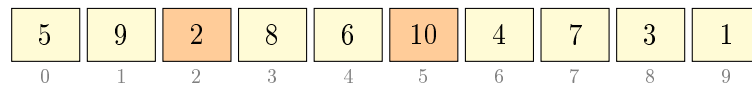
Schritt 5: $a[3]=7$ und $a[7]=8$: Tausche



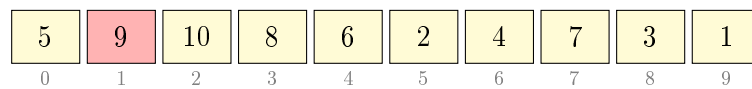
Schritt 6: Heapify ab Index 2 (Wert 2)



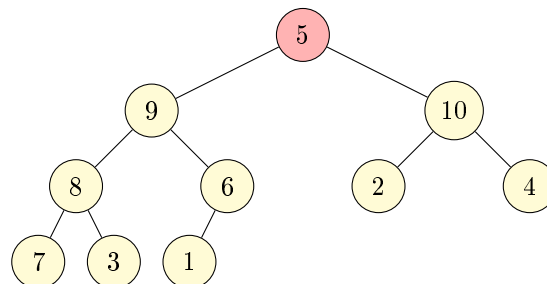
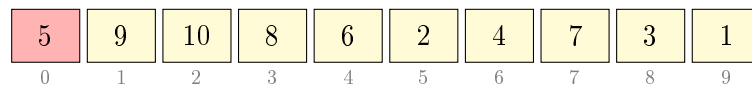
Schritt 7: $a[2]=2$ und $a[5]=10$: Tausche



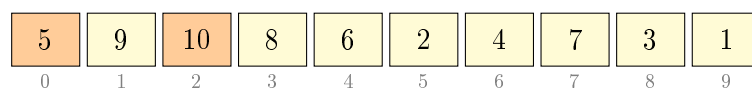
Schritt 8: Heapify ab Index 1 (Wert 9)



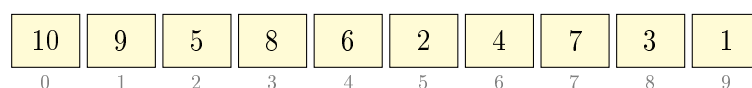
Schritt 9: Heapify ab Index 0 (Wert 5)

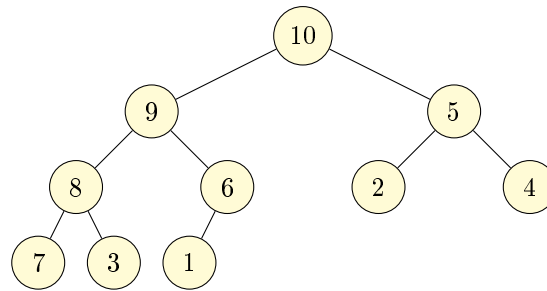


Schritt 10: $a[0]=5$ und $a[2]=10$: Tausche



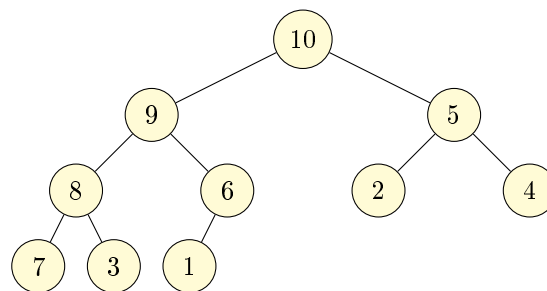
Max-Heap fertig





Antworten:

1. Max-Heap in Array-Darstellung: [10, 9, 5, 8, 6, 2, 4, 7, 3, 1]
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=10 \geq a[1]=9$ (linkes Kind) ✓
- $a[0]=10 \geq a[2]=5$ (rechtes Kind) ✓
- $a[1]=9 \geq a[3]=8$ (linkes Kind) ✓
- $a[1]=9 \geq a[4]=6$ (rechtes Kind) ✓
- $a[2]=5 \geq a[5]=2$ (linkes Kind) ✓
- $a[2]=5 \geq a[6]=4$ (rechtes Kind) ✓
- $a[3]=8 \geq a[7]=7$ (linkes Kind) ✓
- $a[3]=8 \geq a[8]=3$ (rechtes Kind) ✓
- $a[4]=6 \geq a[9]=1$ (linkes Kind) ✓

Aufgabe 4a – Schwierigkeit: ★★★

Min-Heap mit Duplikaten

Stellen Sie die folgenden Zahlen als **Min-Heap** in Array-Darstellung dar:

3, 1, 4, 1, 5, 9, 2, 6

Methode: Bottom-Up Heapify

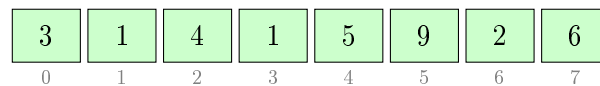
Fragen:

1. Geben Sie den Min-Heap in Array-Darstellung an.
2. Zeichnen Sie den Min-Heap als Binaerbaum.

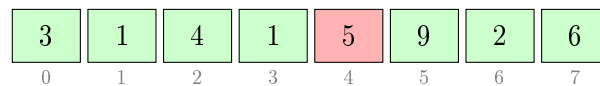
Loesung:

Schritt-fuer-Schritt-Konstruktion:

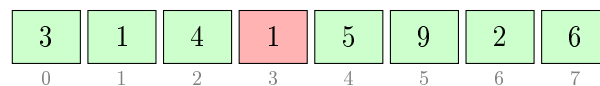
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



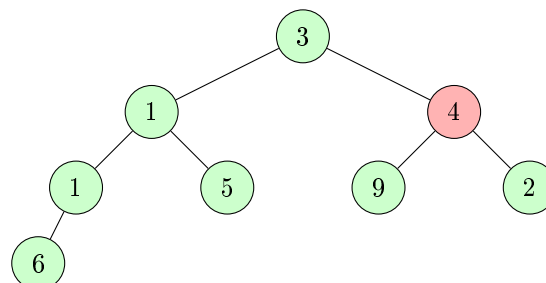
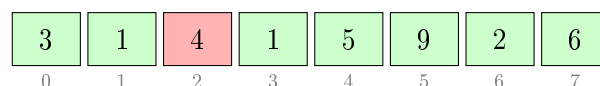
Schritt 1: Heapify ab Index 4 (Wert 5)



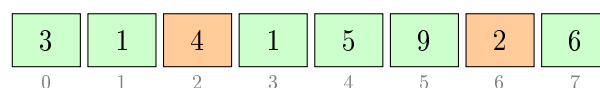
Schritt 2: Heapify ab Index 3 (Wert 1)



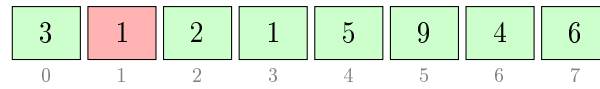
Schritt 3: Heapify ab Index 2 (Wert 4)



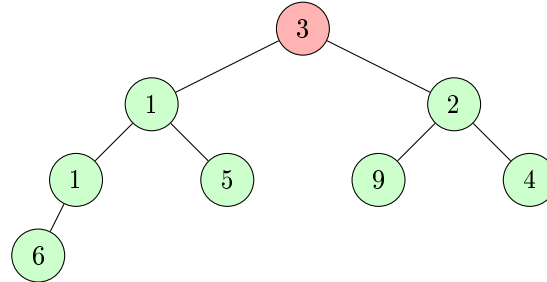
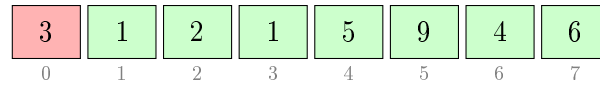
Schritt 4: $a[2]=4$ und $a[6]=2$: Tausche



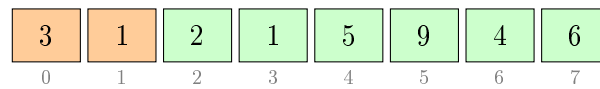
Schritt 5: Heapify ab Index 1 (Wert 1)



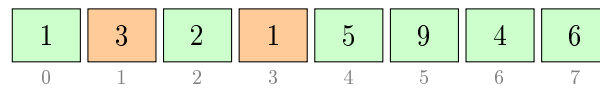
Schritt 6: Heapify ab Index 0 (Wert 3)



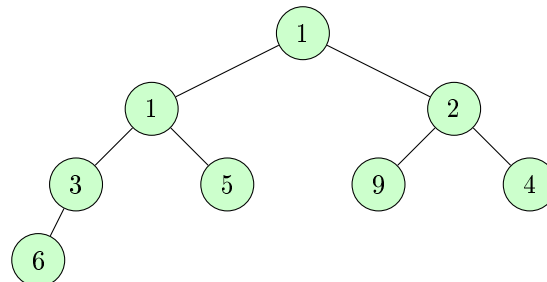
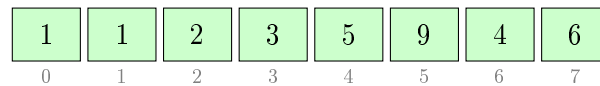
Schritt 7: $a[0]=3$ und $a[1]=1$: Tausche



Schritt 8: $a[1]=3$ und $a[3]=1$: Tausche

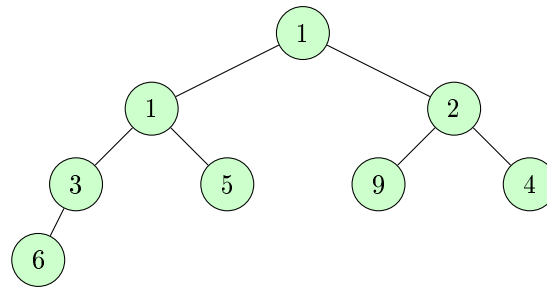


Min-Heap fertig



Antworten:

1. Min-Heap in Array-Darstellung: [1, 1, 2, 3, 5, 9, 4, 6]
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=1 \leq a[1]=1$ (linkes Kind) ✓
 - $a[0]=1 \leq a[2]=2$ (rechtes Kind) ✓
 - $a[1]=1 \leq a[3]=3$ (linkes Kind) ✓
 - $a[1]=1 \leq a[4]=5$ (rechtes Kind) ✓
 - $a[2]=2 \leq a[5]=9$ (linkes Kind) ✓
 - $a[2]=2 \leq a[6]=4$ (rechtes Kind) ✓
 - $a[3]=3 \leq a[7]=6$ (linkes Kind) ✓
-

Aufgabe 4b – Schwierigkeit: ★★★

Max-Heap mit Duplikaten

Stellen Sie die folgenden Zahlen als **Max-Heap** in Array-Darstellung dar:

5, 5, 3, 8, 3, 1, 8, 2

Methode: Bottom-Up Heapify

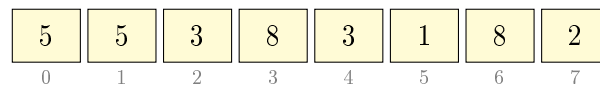
Fragen:

1. Geben Sie den Max-Heap in Array-Darstellung an.
2. Zeichnen Sie den Max-Heap als Binaerbaum.

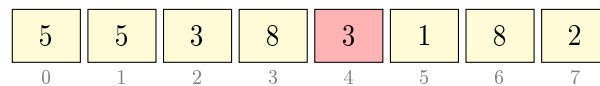
Loesung:

Schritt-fuer-Schritt-Konstruktion:

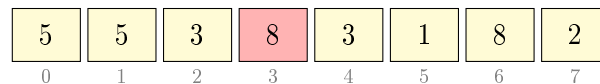
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



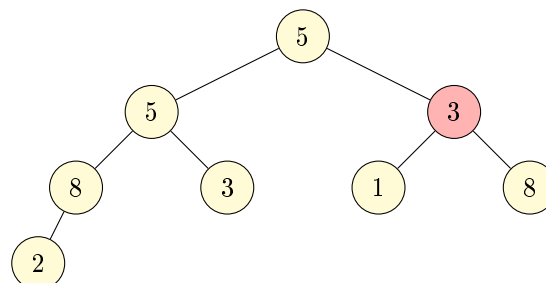
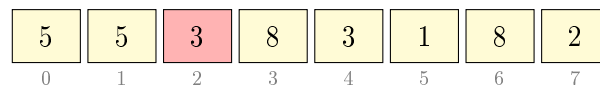
Schritt 1: Heapify ab Index 4 (Wert 3)



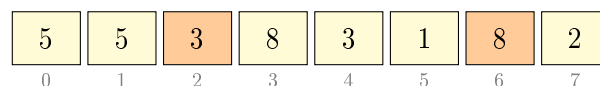
Schritt 2: Heapify ab Index 3 (Wert 8)



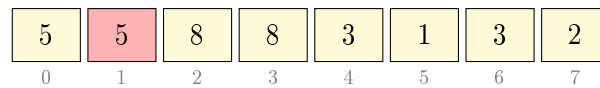
Schritt 3: Heapify ab Index 2 (Wert 3)



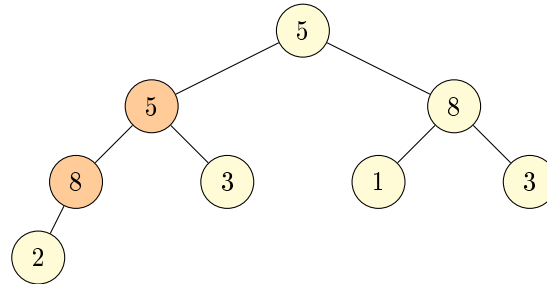
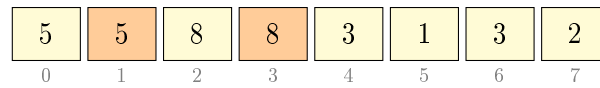
Schritt 4: $a[2]=3$ und $a[6]=8$: Tausche



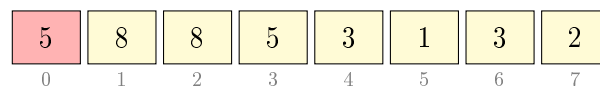
Schritt 5: Heapify ab Index 1 (Wert 5)



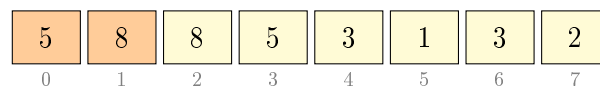
Schritt 6: $a[1]=5$ und $a[3]=8$: Tausche



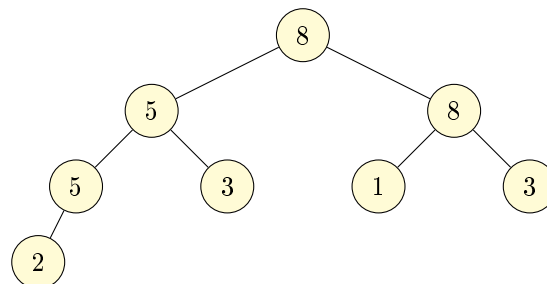
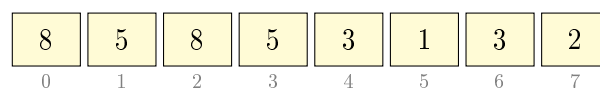
Schritt 7: Heapify ab Index 0 (Wert 5)



Schritt 8: $a[0]=5$ und $a[1]=8$: Tausche

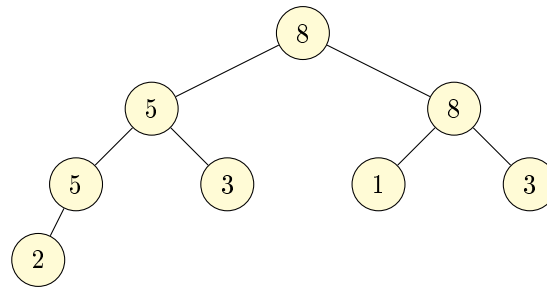


Max-Heap fertig



Antworten:

1. Max-Heap in Array-Darstellung: **[8, 5, 8, 5, 3, 1, 3, 2]**
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=8 \geq a[1]=5$ (linkes Kind) ✓
 - $a[0]=8 \geq a[2]=8$ (rechtes Kind) ✓
 - $a[1]=5 \geq a[3]=5$ (linkes Kind) ✓
 - $a[1]=5 \geq a[4]=3$ (rechtes Kind) ✓
 - $a[2]=8 \geq a[5]=1$ (linkes Kind) ✓
 - $a[2]=8 \geq a[6]=3$ (rechtes Kind) ✓
 - $a[3]=5 \geq a[7]=2$ (linkes Kind) ✓
-

Aufgabe 5a – Schwierigkeit: ★★★★★

Min-Heap aus sortierter absteigender Folge

Stellen Sie die folgenden Zahlen als **Min-Heap** in Array-Darstellung dar:

10, 9, 8, 7, 6, 5, 4, 3, 2, 1

Methode: Bottom-Up Heapify

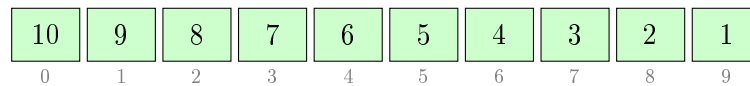
Fragen:

1. Geben Sie den Min-Heap in Array-Darstellung an.
2. Zeichnen Sie den Min-Heap als Binaerbaum.

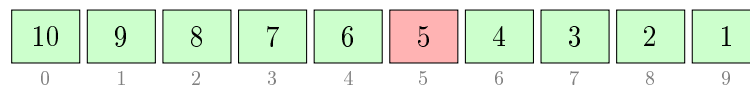
Loesung:

Schritt-fuer-Schritt-Konstruktion:

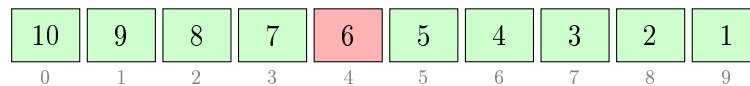
Schritt 0: Ausgangszustand (als Binaerbaum interpretiert)



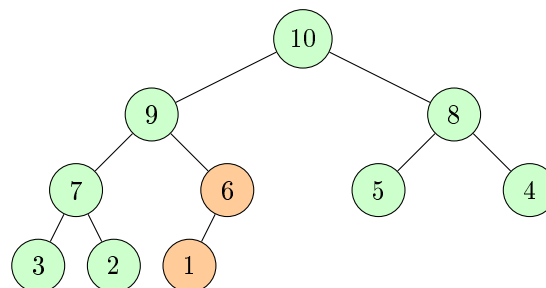
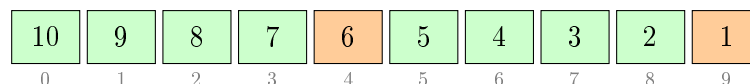
Schritt 1: Heapify ab Index 5 (Wert 5)



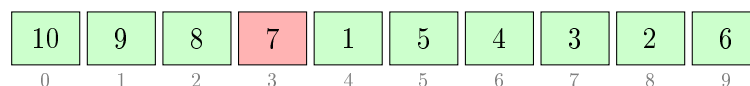
Schritt 2: Heapify ab Index 4 (Wert 6)



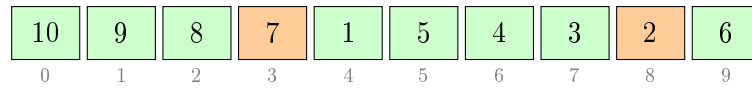
Schritt 3: $a[4]=6$ und $a[9]=1$: Tausche



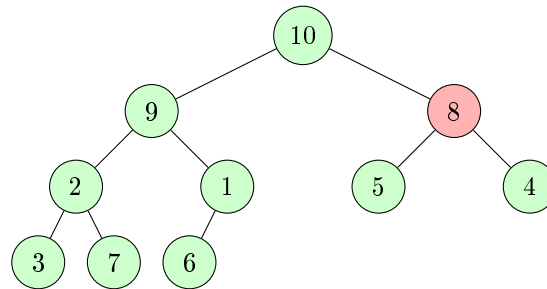
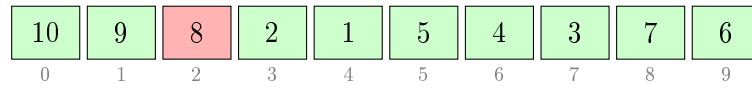
Schritt 4: Heapify ab Index 3 (Wert 7)



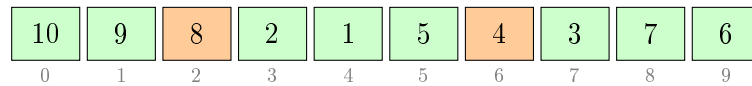
Schritt 5: $a[3]=7$ und $a[8]=2$: Tausche



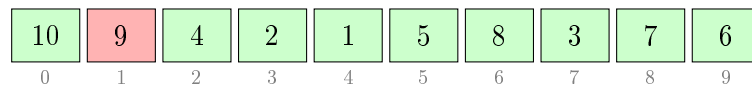
Schritt 6: Heapify ab Index 2 (Wert 8)



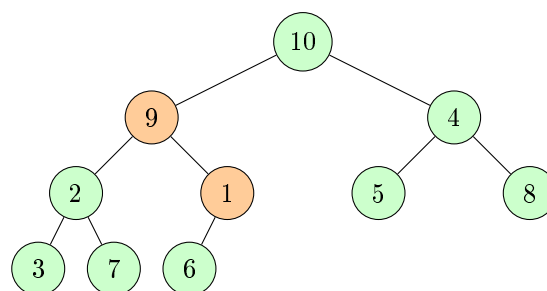
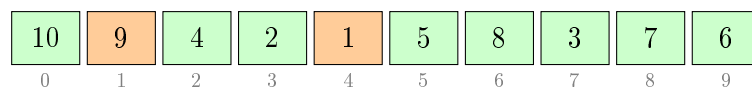
Schritt 7: $a[2]=8$ und $a[6]=4$: Tausche



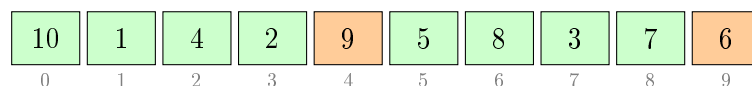
Schritt 8: Heapify ab Index 1 (Wert 9)



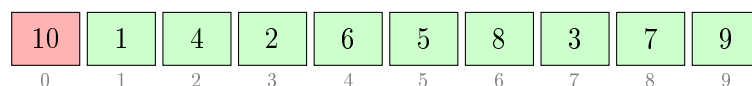
Schritt 9: $a[1]=9$ und $a[4]=1$: Tausche



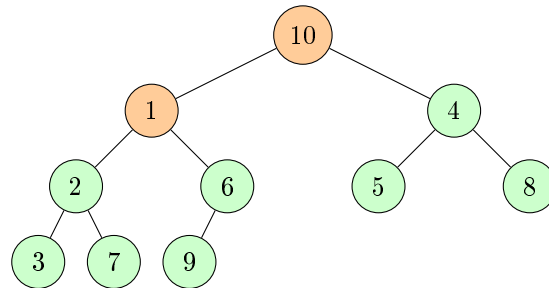
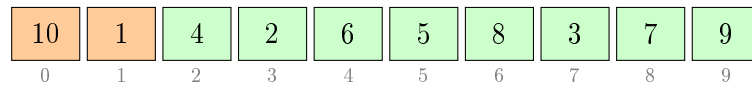
Schritt 10: $a[4]=9$ und $a[9]=6$: Tausche



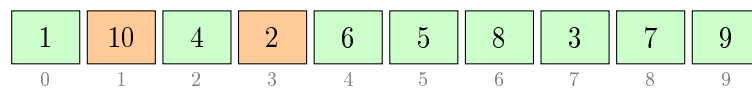
Schritt 11: Heapify ab Index 0 (Wert 10)



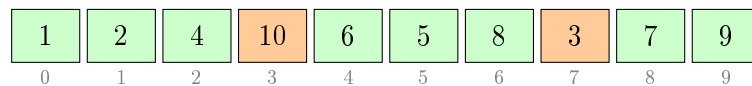
Schritt 12: $a[0]=10$ und $a[1]=1$: Tausche



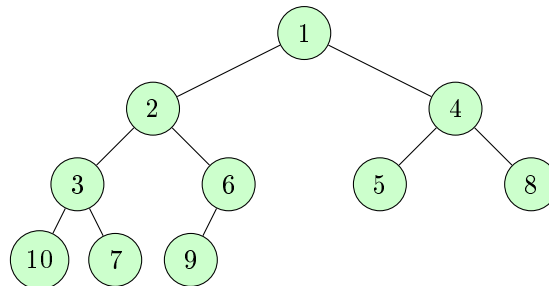
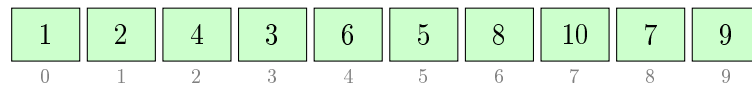
Schritt 13: $a[1]=10$ und $a[3]=2$: Tausche



Schritt 14: $a[3]=10$ und $a[7]=3$: Tausche



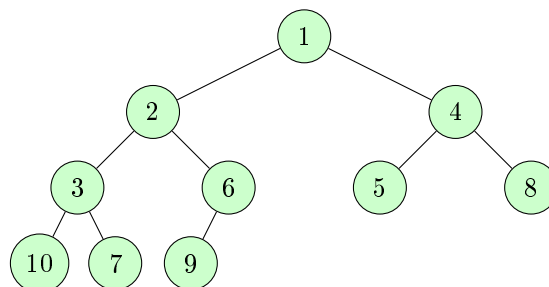
Min-Heap fertig



Antworten:

1. Min-Heap in Array-Darstellung: **[1, 2, 4, 3, 6, 5, 8, 10, 7, 9]**

2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=1 \leq a[1]=2$ (linkes Kind) ✓
 - $a[0]=1 \leq a[2]=4$ (rechtes Kind) ✓
 - $a[1]=2 \leq a[3]=3$ (linkes Kind) ✓
 - $a[1]=2 \leq a[4]=6$ (rechtes Kind) ✓
 - $a[2]=4 \leq a[5]=5$ (linkes Kind) ✓
 - $a[2]=4 \leq a[6]=8$ (rechtes Kind) ✓
 - $a[3]=3 \leq a[7]=10$ (linkes Kind) ✓
 - $a[3]=3 \leq a[8]=7$ (rechtes Kind) ✓
 - $a[4]=6 \leq a[9]=9$ (linkes Kind) ✓
-

Aufgabe 5b – Schwierigkeit: ★★★★★

Max-Heap aus sortierter aufsteigender Folge

Stellen Sie die folgenden Zahlen als **Max-Heap** in Array-Darstellung dar:

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Methode: Sukzessives Einfuegen mit Sift-Up

Fragen:

1. Geben Sie den Max-Heap in Array-Darstellung an.
2. Zeichnen Sie den Max-Heap als Binaerbaum.

Loesung:

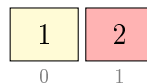
Schritt-fuer-Schritt-Konstruktion:

Schritt 0: Leerer Heap

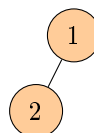
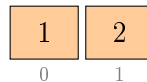
Schritt 1: Fuege 1 am Ende ein (Index 0)



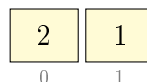
Schritt 2: Fuege 2 am Ende ein (Index 1)



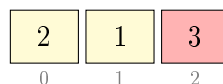
Schritt 3: $a[1]=2 > a[0]=1$ (Eltern): Tausche



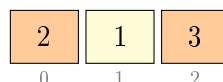
Schritt 4: 2 ist an Wurzel angekommen

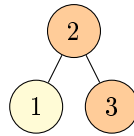


Schritt 5: Fuege 3 am Ende ein (Index 2)

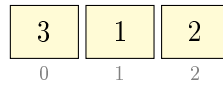


Schritt 6: $a[2]=3 > a[0]=2$ (Eltern): Tausche

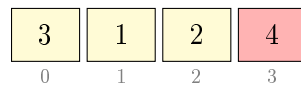




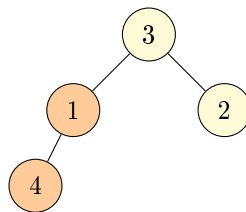
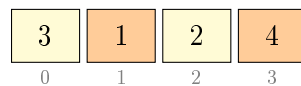
Schritt 7: 3 ist an Wurzel angekommen



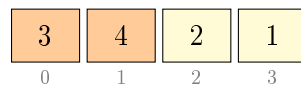
Schritt 8: Füge 4 am Ende ein (Index 3)



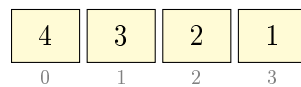
Schritt 9: $a[3]=4 > a[1]=1$ (Eltern): Tausche



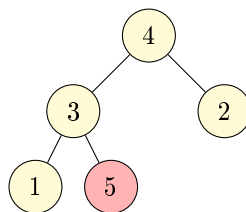
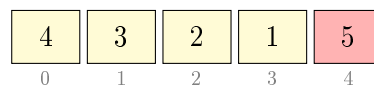
Schritt 10: $a[1]=4 > a[0]=3$ (Eltern): Tausche



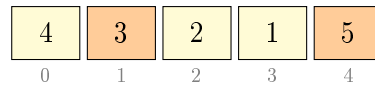
Schritt 11: 4 ist an Wurzel angekommen



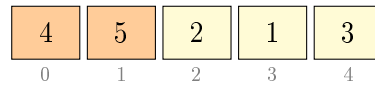
Schritt 12: Füge 5 am Ende ein (Index 4)



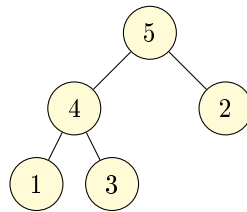
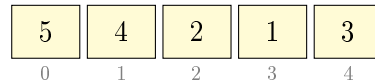
Schritt 13: $a[4]=5 > a[1]=3$ (Eltern): Tausche



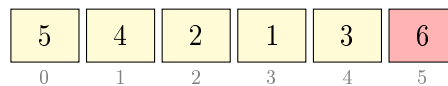
Schritt 14: $a[1]=5 > a[0]=4$ (Eltern): Tausche



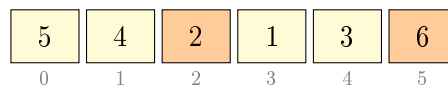
Schritt 15: 5 ist an Wurzel angekommen



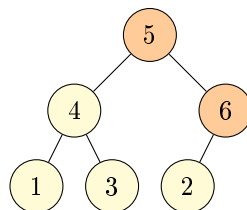
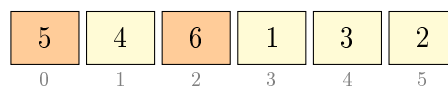
Schritt 16: Füge 6 am Ende ein (Index 5)



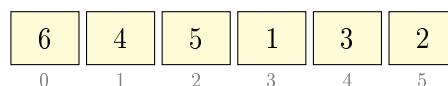
Schritt 17: $a[5]=6 > a[2]=2$ (Eltern): Tausche



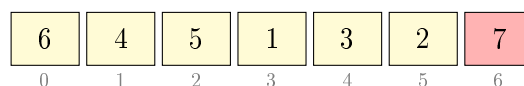
Schritt 18: $a[2]=6 > a[0]=5$ (Eltern): Tausche



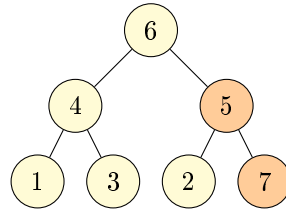
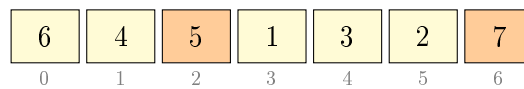
Schritt 19: 6 ist an Wurzel angekommen



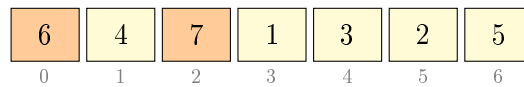
Schritt 20: Füge 7 am Ende ein (Index 6)



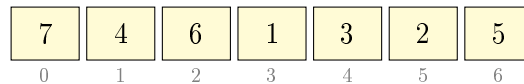
Schritt 21: $a[6]=7 > a[2]=5$ (Eltern): Tausche



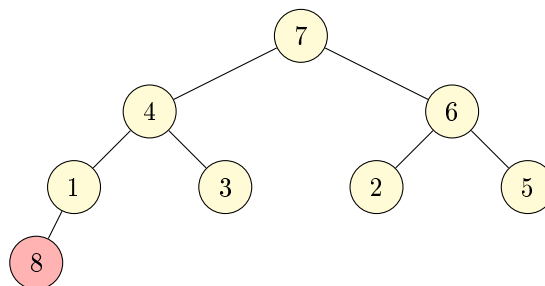
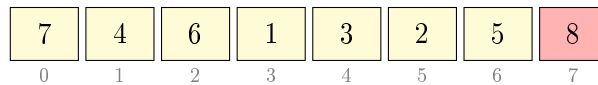
Schritt 22: $a[2]=7 > a[0]=6$ (Eltern): Tausche



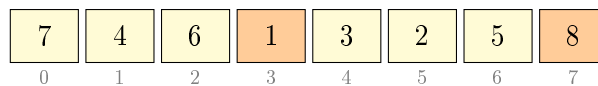
Schritt 23: 7 ist an Wurzel angekommen



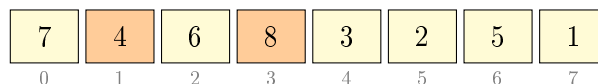
Schritt 24: Füge 8 am Ende ein (Index 7)



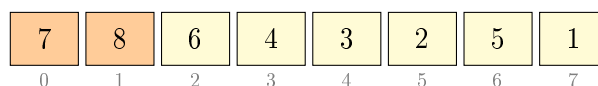
Schritt 25: $a[7]=8 > a[3]=1$ (Eltern): Tausche

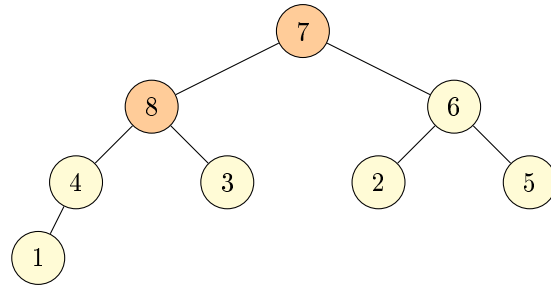


Schritt 26: $a[3]=8 > a[1]=4$ (Eltern): Tausche

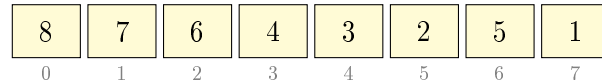


Schritt 27: $a[1]=8 > a[0]=7$ (Eltern): Tausche

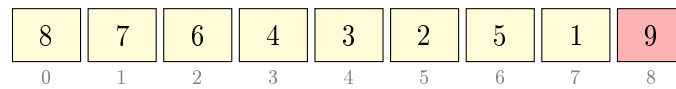




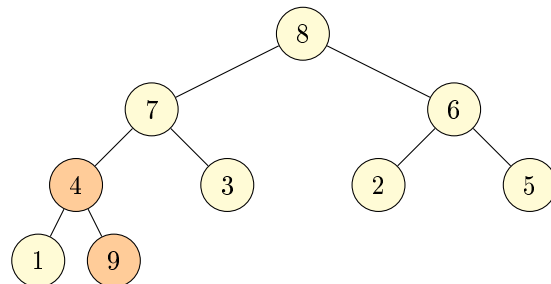
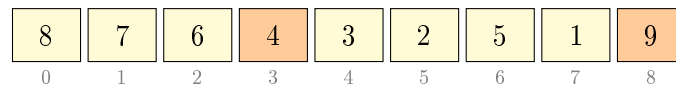
Schritt 28: 8 ist an Wurzel angekommen



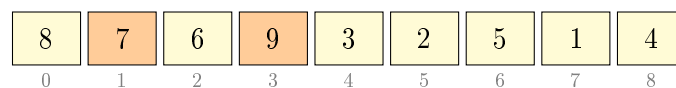
Schritt 29: Füge 9 am Ende ein (Index 8)



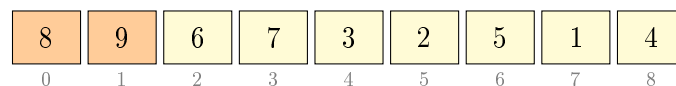
Schritt 30: $a[8]=9 > a[3]=4$ (Eltern): Tausche



Schritt 31: $a[3]=9 > a[1]=7$ (Eltern): Tausche

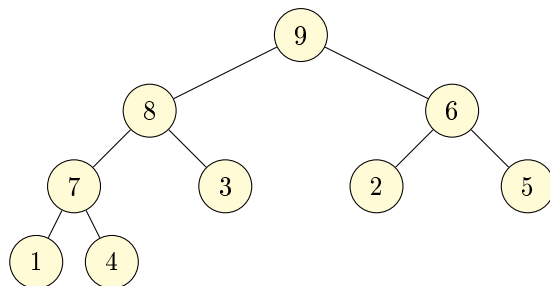


Schritt 32: $a[1]=9 > a[0]=8$ (Eltern): Tausche

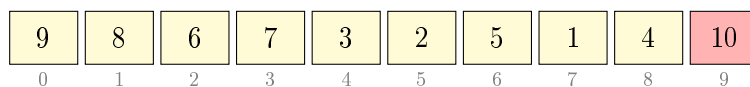


Schritt 33: 9 ist an Wurzel angekommen

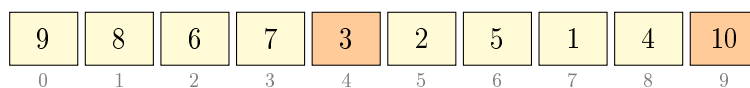




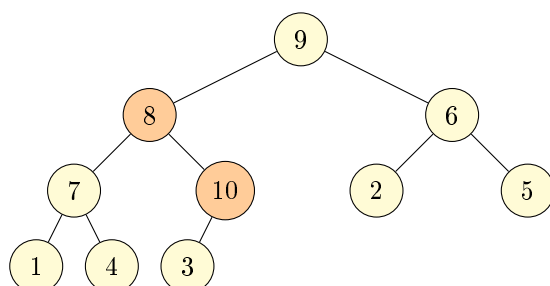
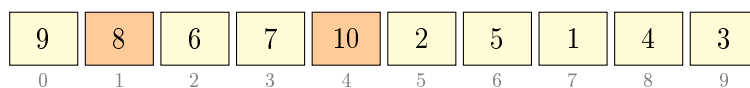
Schritt 34: Füge 10 am Ende ein (Index 9)



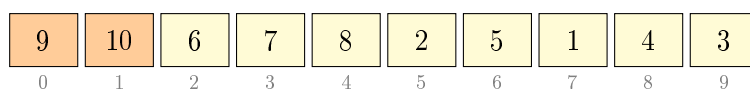
Schritt 35: $a[9]=10 > a[4]=3$ (Eltern): Tausche



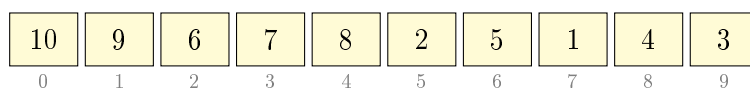
Schritt 36: $a[4]=10 > a[1]=8$ (Eltern): Tausche



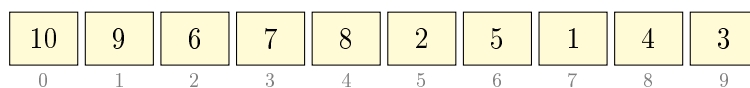
Schritt 37: $a[1]=10 > a[0]=9$ (Eltern): Tausche

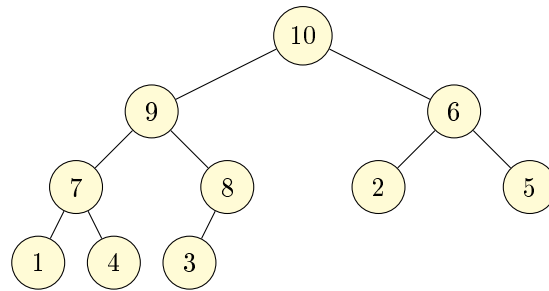


Schritt 38: 10 ist an Wurzel angekommen



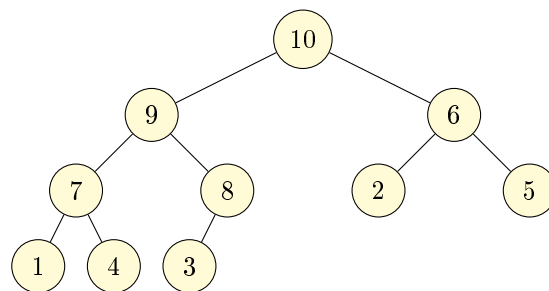
Max-Heap fertig





Antworten:

1. Max-Heap in Array-Darstellung: [10, 9, 6, 7, 8, 2, 5, 1, 4, 3]
2. Baumdarstellung:



Verifikation der Heap-Eigenschaft:

- $a[0]=10 \geq a[1]=9$ (linkes Kind) ✓
 - $a[0]=10 \geq a[2]=6$ (rechtes Kind) ✓
 - $a[1]=9 \geq a[3]=7$ (linkes Kind) ✓
 - $a[1]=9 \geq a[4]=8$ (rechtes Kind) ✓
 - $a[2]=6 \geq a[5]=2$ (linkes Kind) ✓
 - $a[2]=6 \geq a[6]=5$ (rechtes Kind) ✓
 - $a[3]=7 \geq a[7]=1$ (linkes Kind) ✓
 - $a[3]=7 \geq a[8]=4$ (rechtes Kind) ✓
 - $a[4]=8 \geq a[9]=3$ (linkes Kind) ✓
-